Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/14/2007
PI Name:	Packard, Richard E. Ph.D.		
Project Title:	Development of High Sensitivity Sensors of Absolute Rotation using Quantum Phase Coherence In Superfluid Helium		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	07/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: received NCE to 7/31/2007 per D. Strayer/JPL (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2876		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We will enhance the sensitivity of a superfluid gyroscope based on a dc-SQUID device developed in an earlier grant. We hope to determine the practical limitation of this technology. Task Objective: 1. Determine the current-phase characteristic of He3 weak links at elevated ambient pressure in order to find the optimal operating point for a superfluid gyroscope. 2. Enhance the sensitivity of the superfluid gyroscope by a factor of 10-100 over pre-existing devices. The superfluid gyroscope can be the most sensitive sensor of rotation extant. Only a prototype device has been		

	demonstrated previously. If the superfluid gyroscope can reach its ultimate theoretical sensitivity it will enhance measurements of the Earth's rotation vector, which ultimately provides a limitation on the GPS navigation system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The superfluid gyroscope is intended as a new sensitive tool for seismic studies and geodesy investigations. The latter may lead to enhanced precision of the GPS navigation system. The gyroscope is a specific use of a more general device named a SHeQUID, an acronym for Superfluid Helium Interference Device. The SHeQUID can be used to probe physical phenomena that are inaccessible to other kinds of probes
Task Progress:	We have succeeded in meeting all goals of our proposal. We developed an entirely new kind of superfluid gyroscope based on He4 rather than on He3. Our new device operates at 2000 times warmer temperature than the originally conceived He3 device. This enables the technology to move from the laboratory into the real world. We have used the SHeQUID (Superfluid Helium Quantum Interference Device) to detect the rotation of the Earth and to monitor superfluid flow in a small pipe. This in turn enabled us to linearize the device giving it high dynamic range as well as high sensitivity.
Bibliography Type:	Description: (Last Updated: 09/12/2008)
Articles in Peer-reviewed Journals	Sato Y, Hoskinson E, Packard R. "Transition from synchronous to asynchronous superfluid phase slippage in an aperture array." Physical Review B. 2006 Oct;74:(14)144502. http://dx.doi.org/10.1103/PhysRevB.74.144502 , Oct-2006
Articles in Peer-reviewed Journals	Sato Y, Joshi A, Packard R. "Direct measurement of quantum phase gradients in superfluid 4He flow." Phys Rev Lett. 2007 May 11;98(19):195302. PMID: 17677625 , May-2007
Articles in Peer-reviewed Journals	Sato Y, Joshi A, Packard R. "Observation of an extrinsic critical velocity using matter wave interferometry." Physical Review B 2007 Aug;76(5):052505. http://dx.doi.org/10.1103/PhysRevB.76.052505 , Aug-2007
Articles in Peer-reviewed Journals	Sato Y, Joshi A, Packard R. "Flux locking a superfluid interferometer." Appl Phys Lett. 2007 Aug 13;91(7):074107. http://dx.doi.org/10.1063/1.2772659 , Aug-2007
Awards	Packard, Richard. "2005 London Award for Low Temperature Physics, July 2005." Jul-2005